SCEC TPV104, strike slip rate at the nine on-fault stations (rows: depth 3 / 7.5 / 12 km; columns: strike)

x2_-12_x3_3 (strike -12 km, depth 3 km)

x2_0_x3_3 (strike 0 km, depth 3 km)

x2_12_x3_3 (strike 12 km, depth 3 km)

x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)

x2_0_x3_7.5 (strike 0 km, depth 7.5 km)

x2_9_x3_7.5 (strike 9 km, depth 7.5 km)

x2_-12_x3_12 (strike -12 km, depth 12 km)

x2_0_x3_12 (strike 0 km, depth 12 km)

x2_12_x3_12 (strike 12 km, depth 12 km)

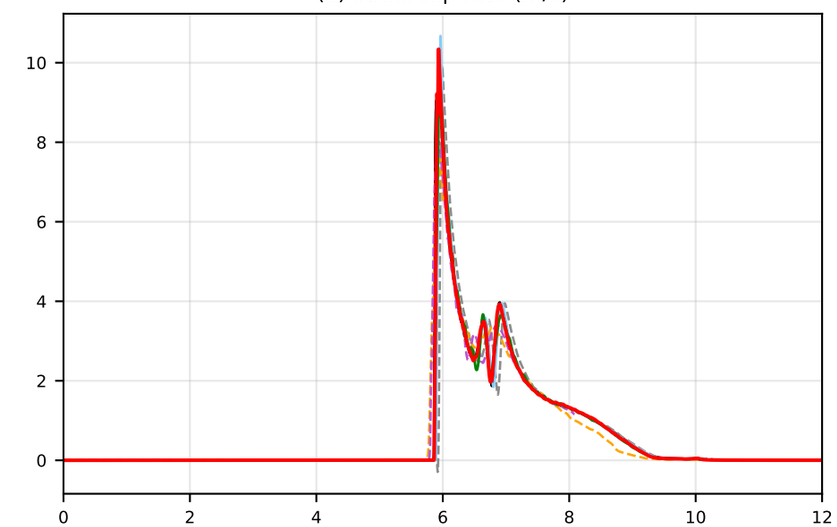
strike slip rate (m/s)

time (s)

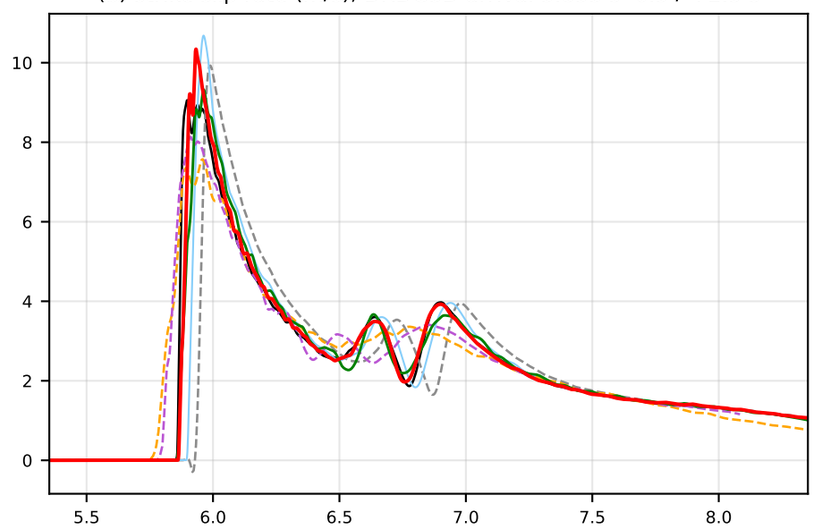
DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_-12_x3_3 (strike -12 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.008 s slip -0.55% peak +13.8%

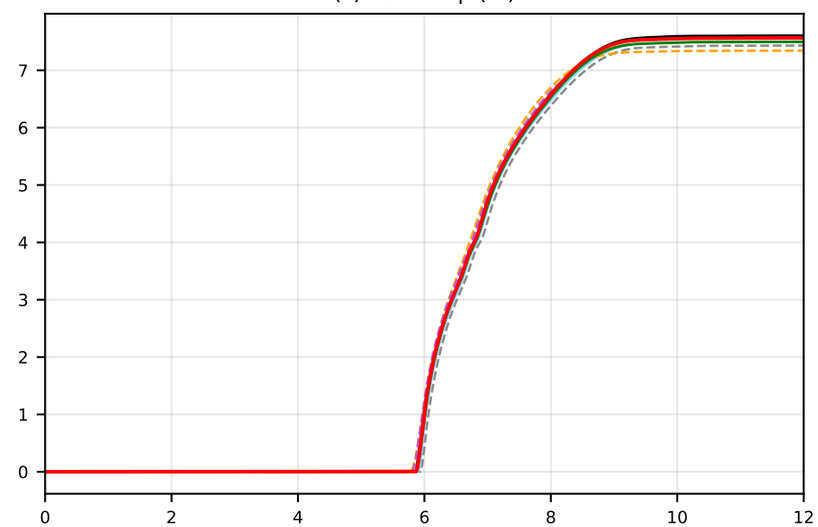
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.853 s -0.5 / +2.5 s

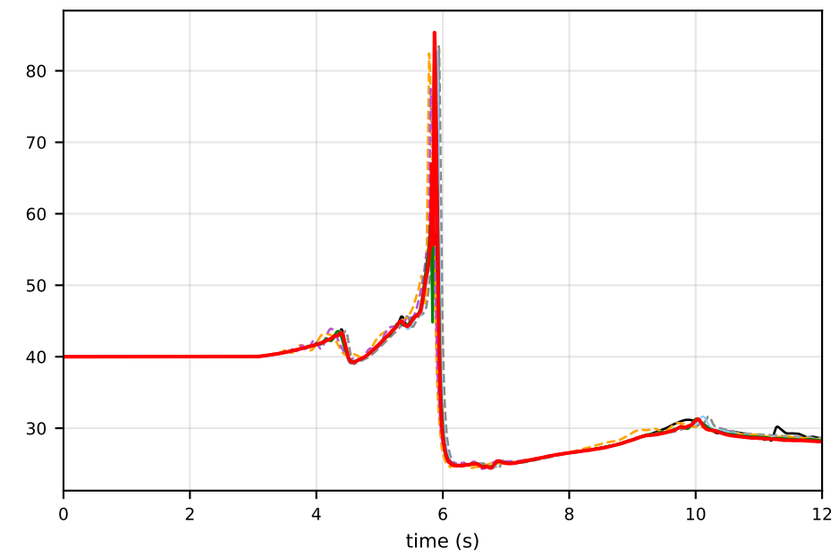


(c) strike slip (m)

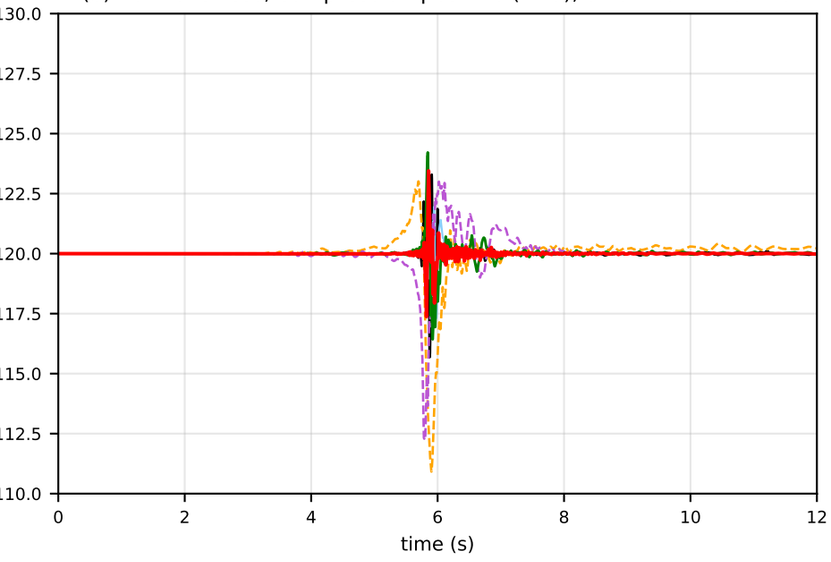


time (s)

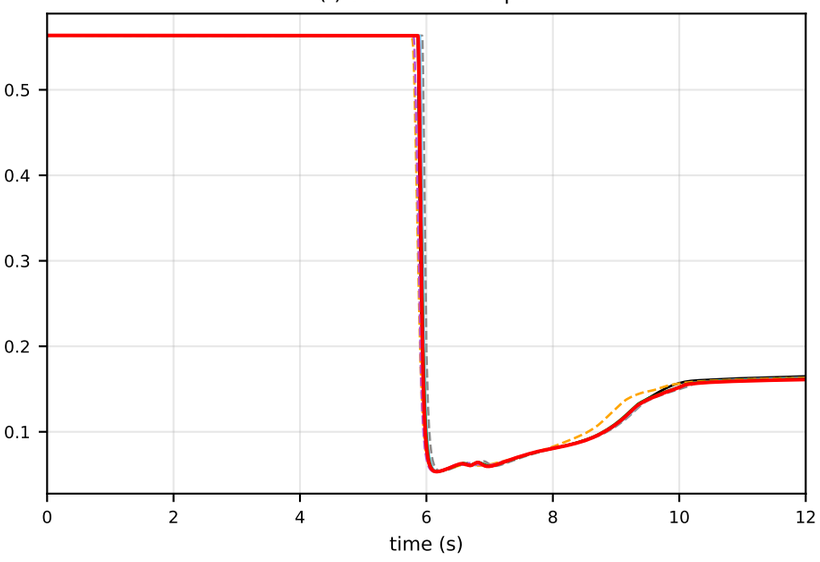
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

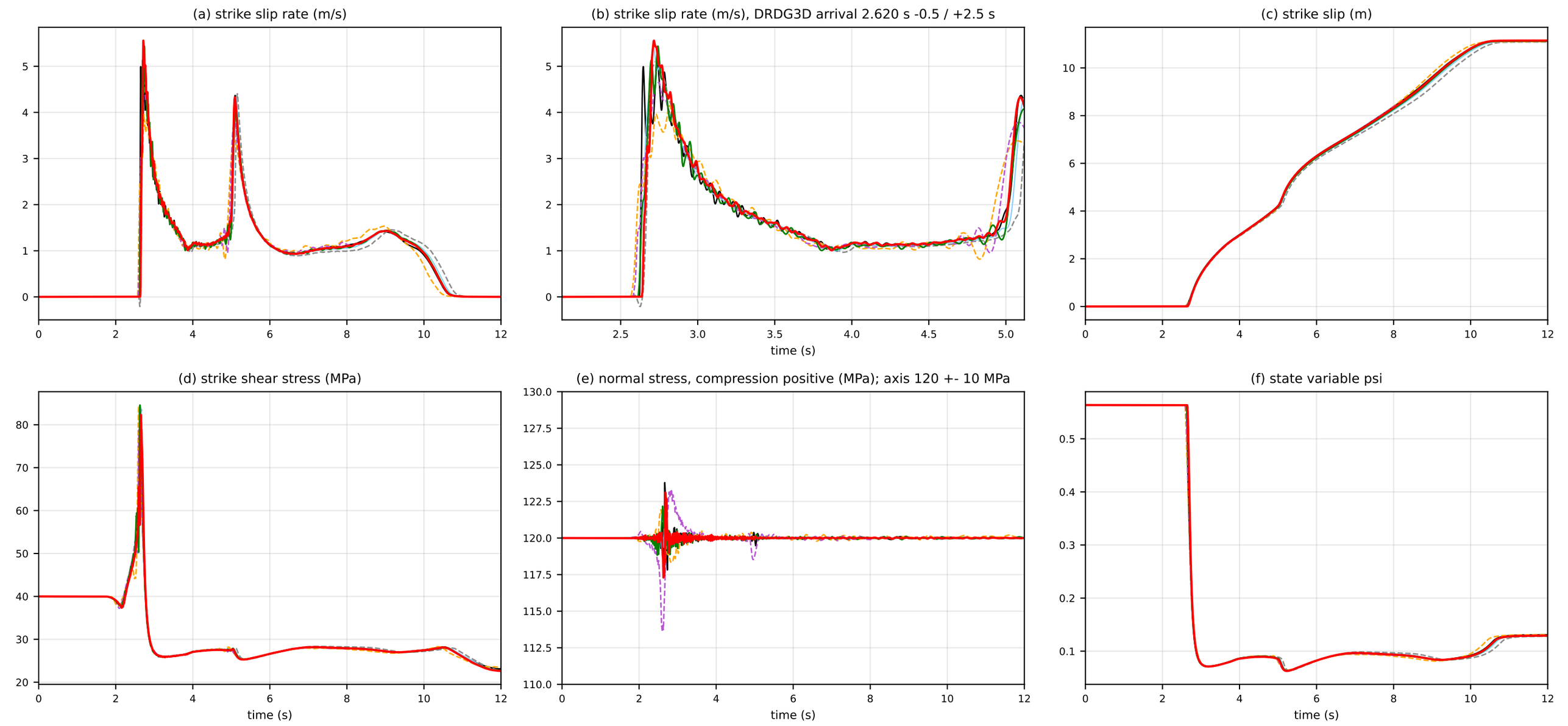


(f) state variable psi



DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_0_x3_3 (strike 0 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.016 s slip +0.07% peak +7.1%



SCEC TPV104 station x2_12_x3_3 (strike 12 km, depth 3 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt -0.035 s slip -0.55% peak +13.2%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 5.854 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); autoscaled: data span 107.6-131.6 MPa

(f) state variable psi

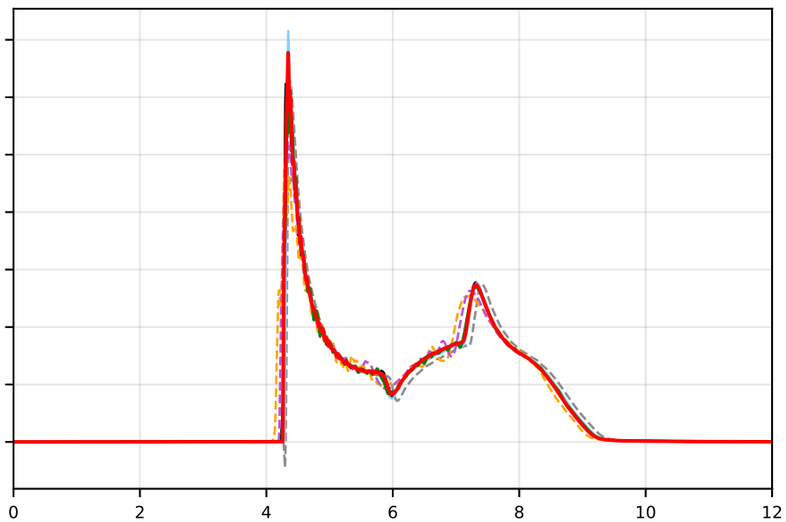
time (s)

time (s)

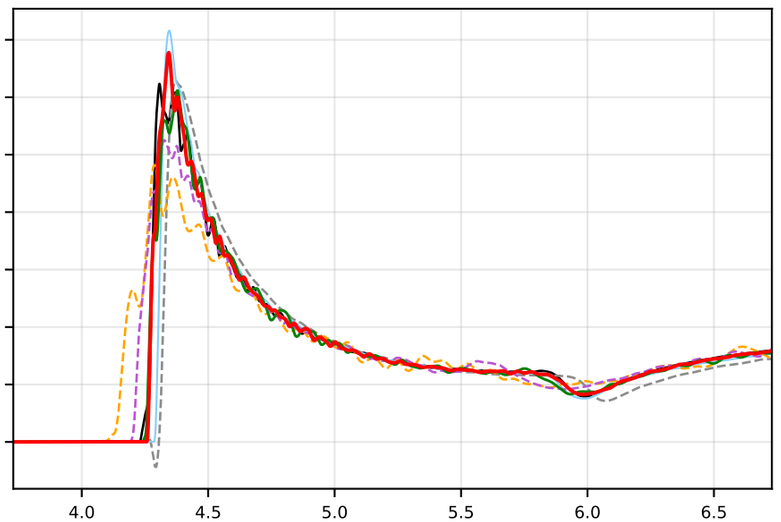
DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_-9_x3_7.5 (strike -9 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.027 s slip -0.28% peak +8.7%

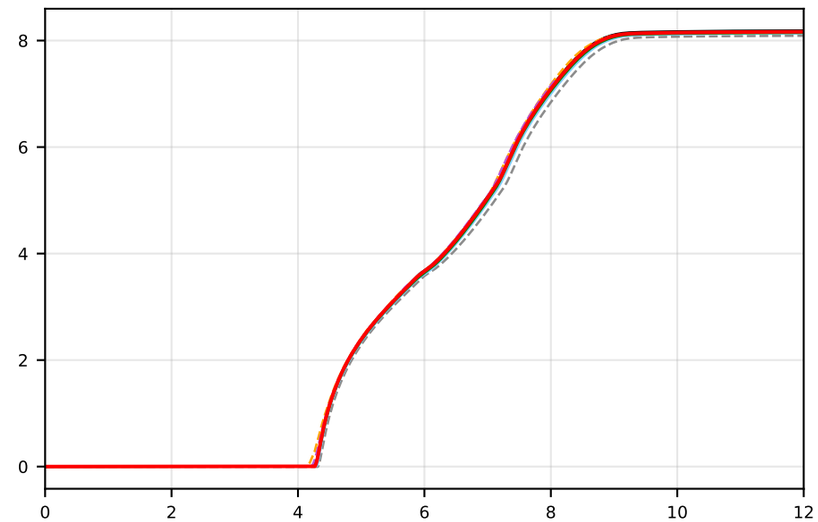
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 4.229 s -0.5 / +2.5 s

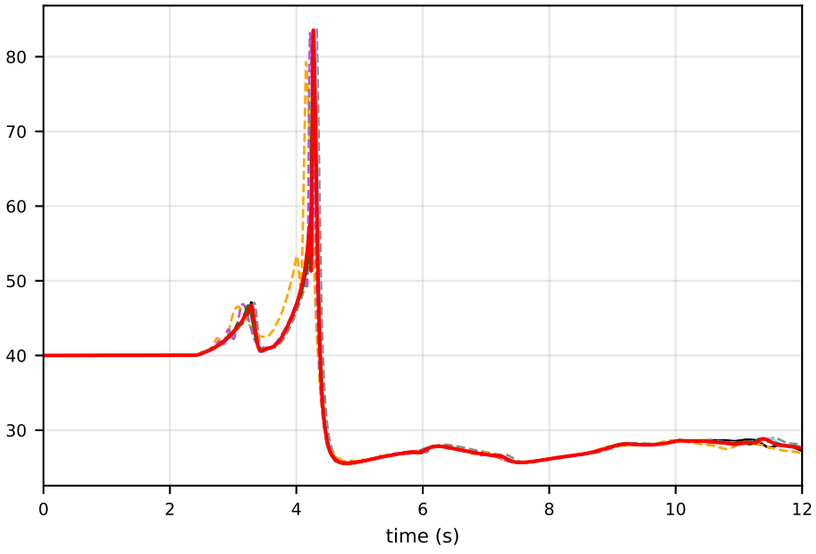


(c) strike slip (m)

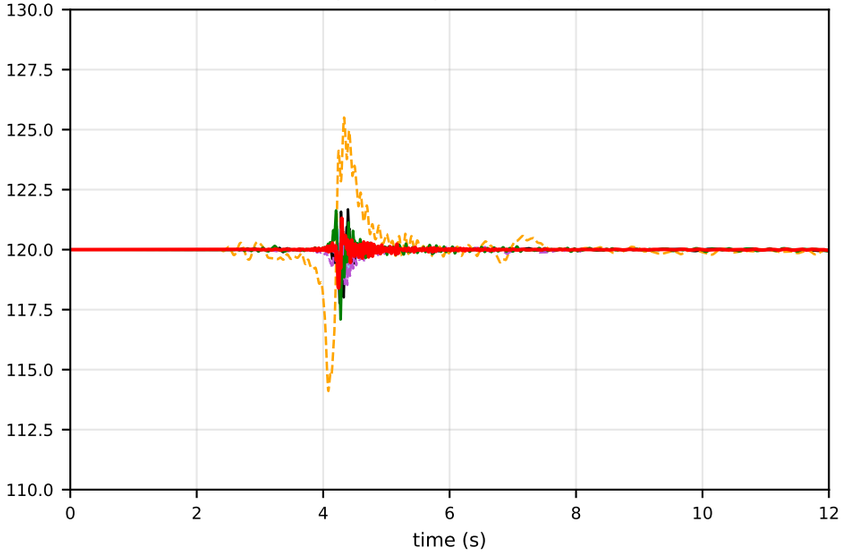


time (s)

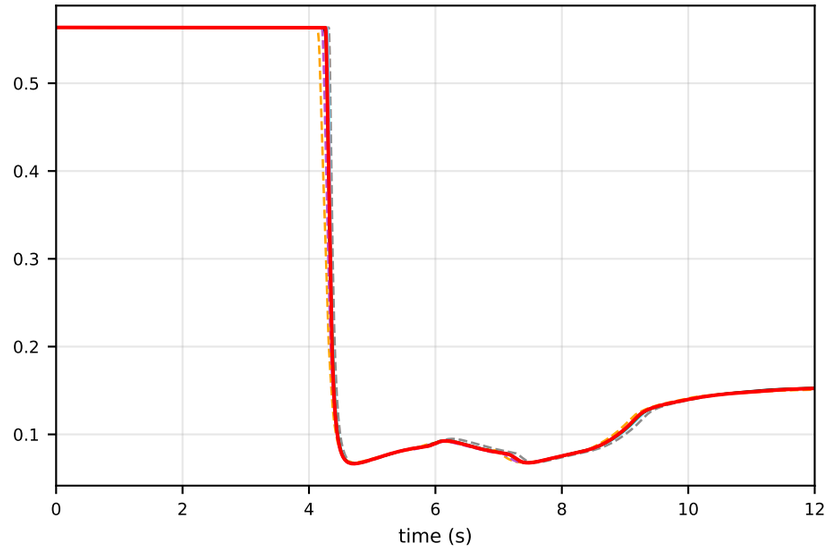
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa



(f) state variable psi



DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_0_x3_7.5 (strike 0 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt -0.003 s slip -0.03% peak +0.3%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 0.574 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); axis 120 +- 10 MPa

(f) state variable psi

time (s)

time (s)

DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_9_x3_7.5 (strike 9 km, depth 7.5 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.028 s slip -0.28% peak +1.9%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 4.229 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); autoscaled: data span 111.3-130.6 MPa

(f) state variable psi

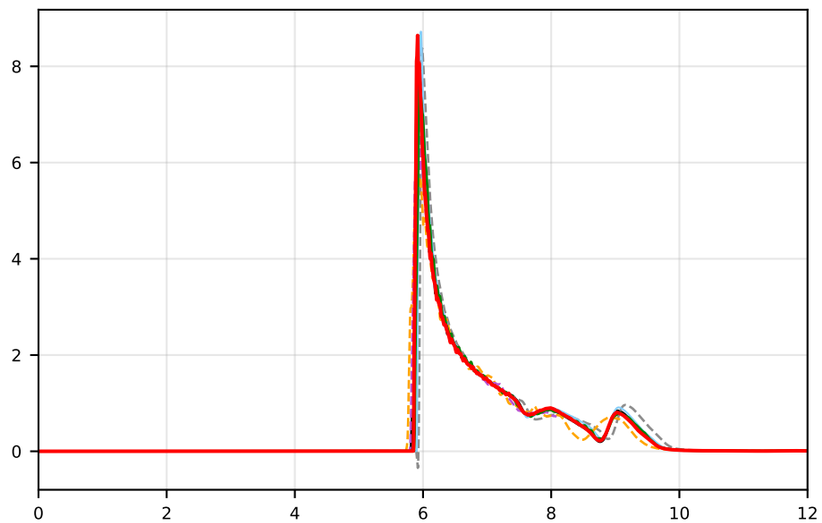
time (s)

time (s)

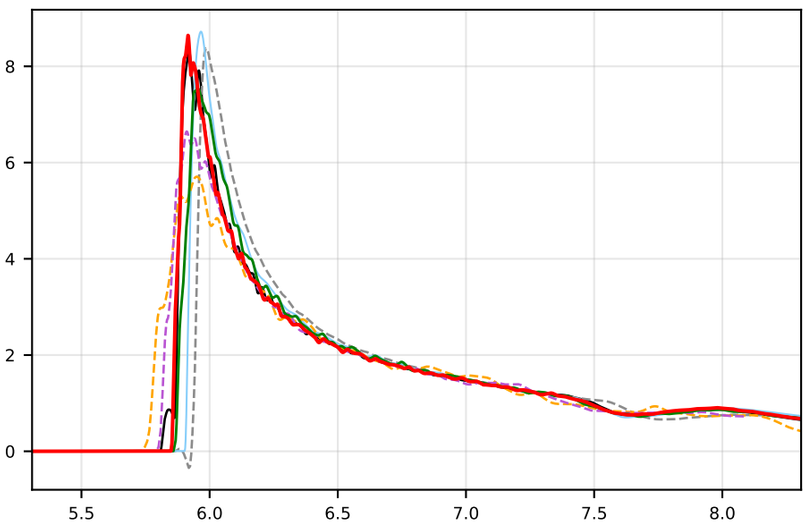
DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_-12_x3_12 (strike -12 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.041 s slip -0.15% peak +3.8%

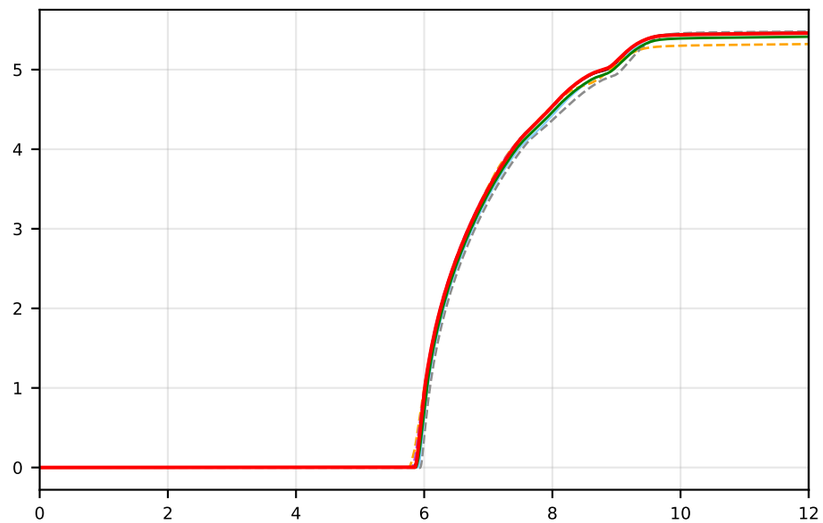
(a) strike slip rate (m/s)



(b) strike slip rate (m/s), DRDG3D arrival 5.807 s -0.5 / +2.5 s

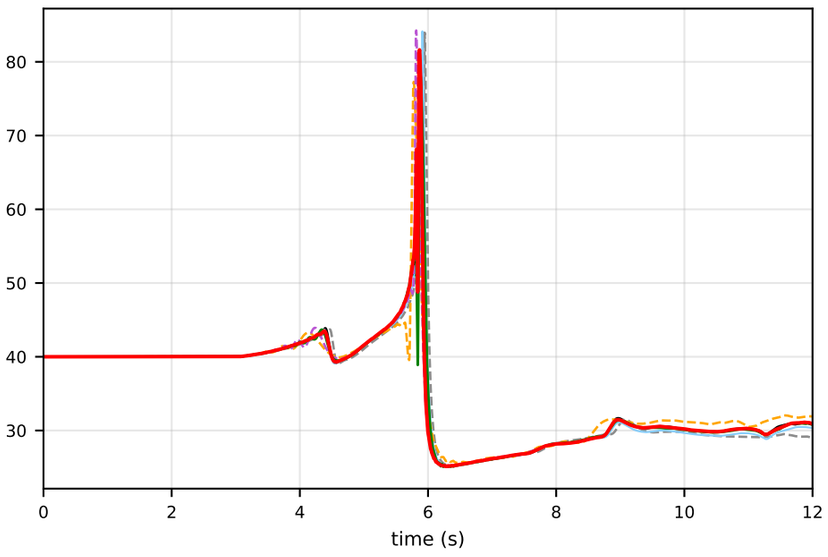


(c) strike slip (m)

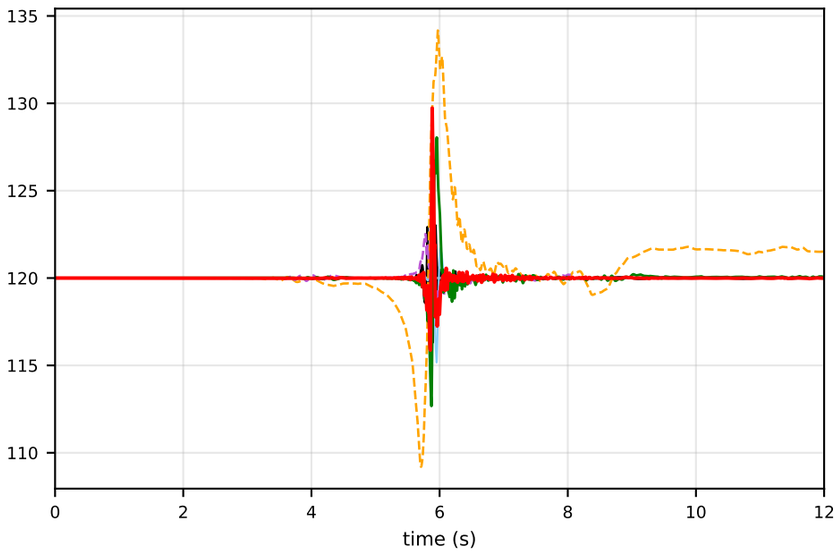


time (s)

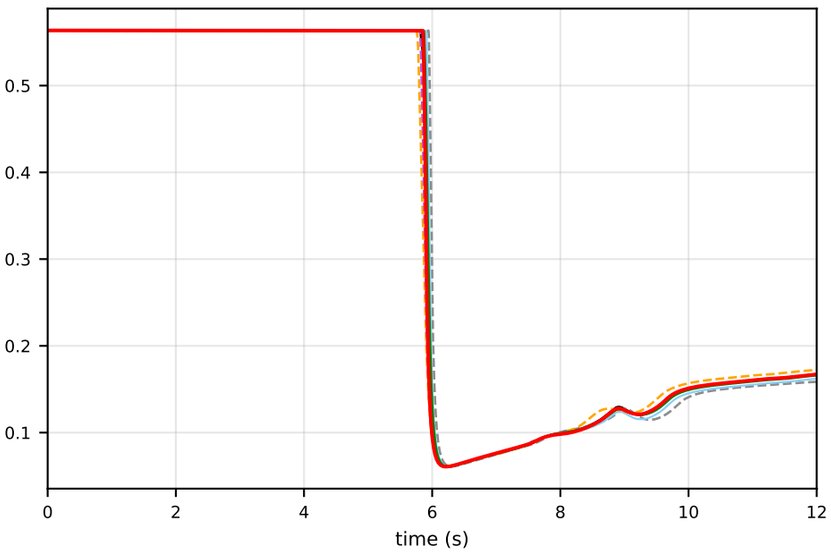
(d) strike shear stress (MPa)



(e) normal stress, compression positive (MPa); autoscaled: data span 109.2-134.2 MPa

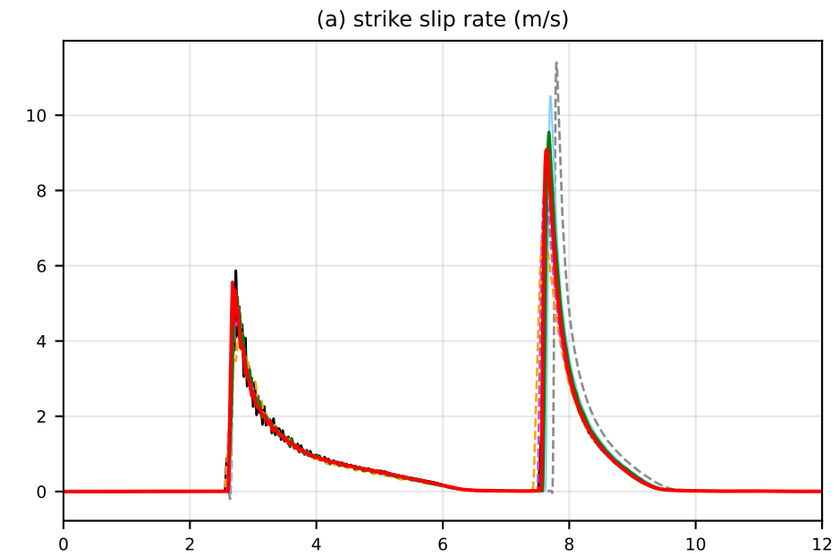


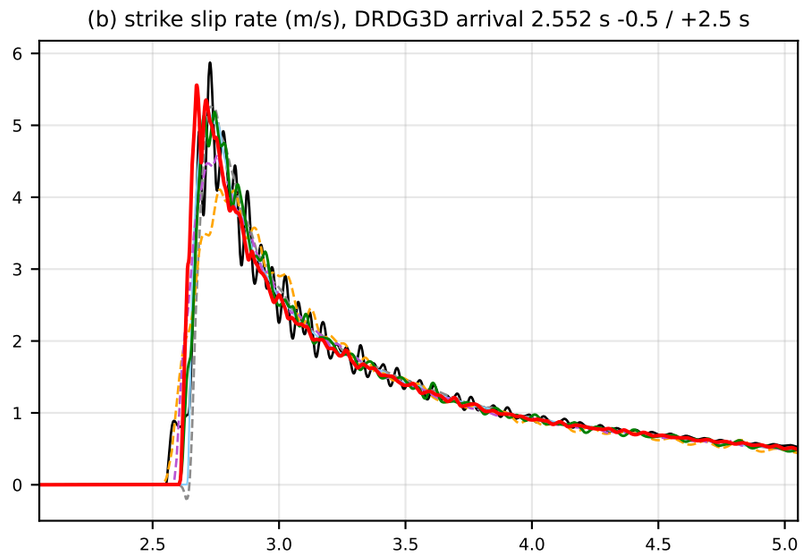
(f) state variable psi

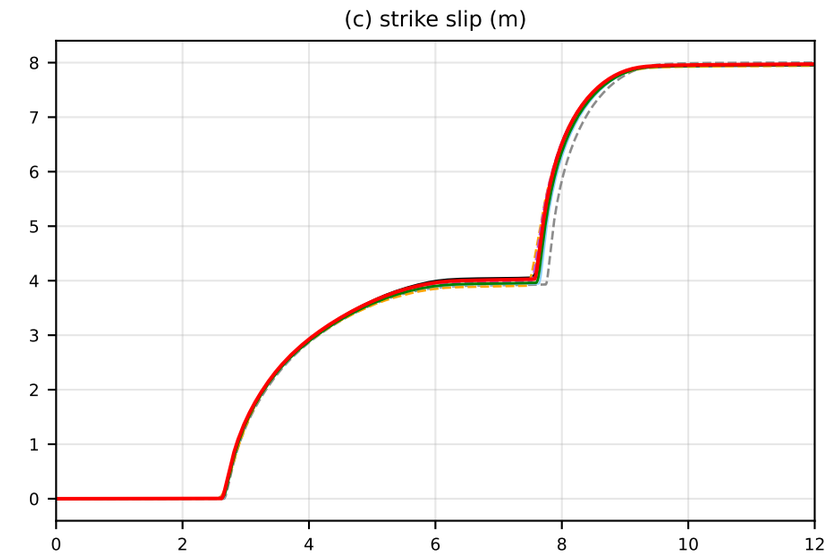


DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

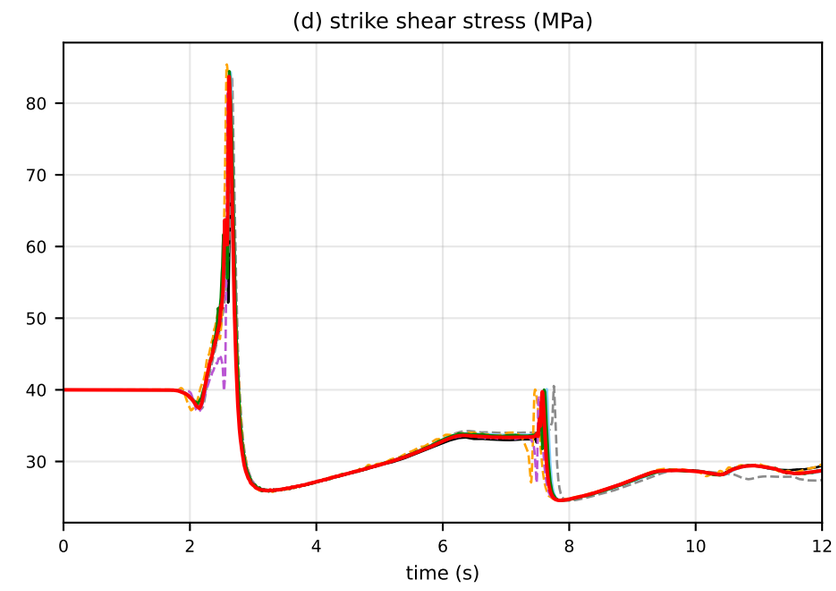
SCEC TPV104 station x2_0_x3_12 (strike 0 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.052 s slip +0.27% peak +6.9%

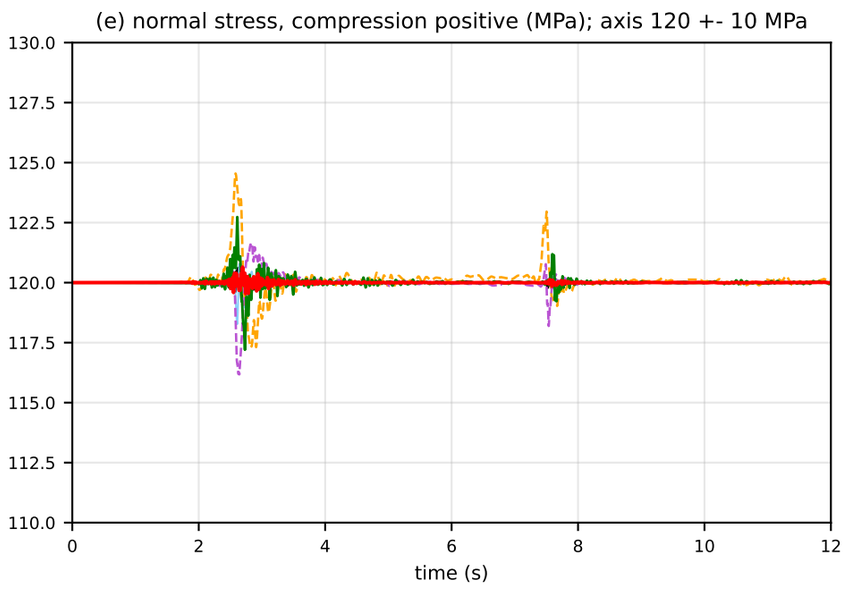


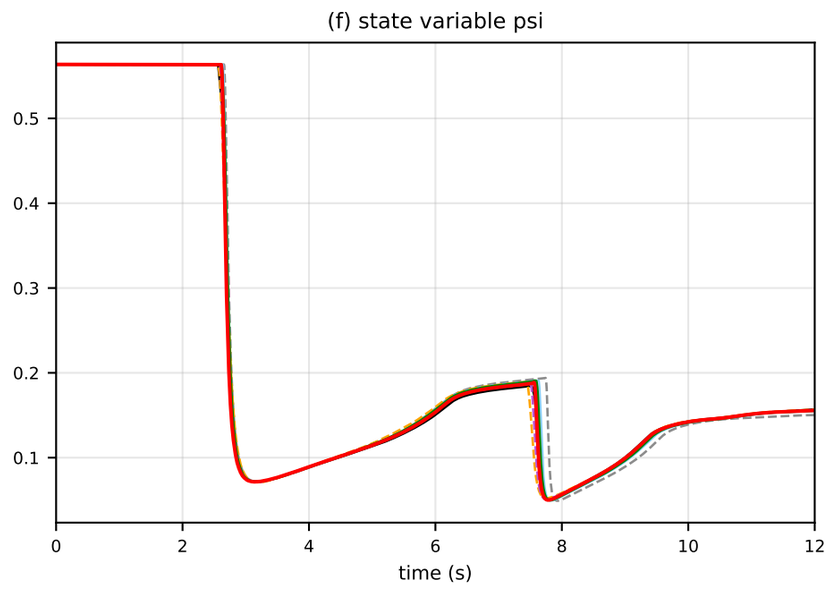




time (s)







DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)

SCEC TPV104 station x2_12_x3_12 (strike 12 km, depth 12 km)
Nitsche p=3 200 m (Vista 1080362) vs DRDG3D: dt +0.042 s slip -0.12% peak -3.5%

(a) strike slip rate (m/s)

(b) strike slip rate (m/s), DRDG3D arrival 5.809 s -0.5 / +2.5 s

(c) strike slip (m)

time (s)

(d) strike shear stress (MPa)

(e) normal stress, compression positive (MPa); autoscaled: data span 106.0-128.4 MPa

(f) state variable psi

time (s)

time (s)

DRDG3D 200 m O4
SeisSol O5
MFEM mixed-flux RK45 p=2 (gold)
Nitsche p=1 200 m (Vista 1075919)
Nitsche p=1 100 m (Vista 1076418, to t = 8.08 s)
Nitsche p=2 200 m (Vista 1075917)
Nitsche p=3 200 m (Vista 1080362)